Foundations of Mathematical Reasoning

Set Theory, Logic, and Combinatorial Thinking

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Duration: 5 months (extendable to 6 months)

Format: **Lectures + Problem-Solving + Assignments**

Course Snapshot: This module introduces rigorous mathematical reasoning through a systematic study of logic, set theory, and combinatorics. It emphasizes proof techniques, conceptual connections, and problem-solving skills required for advanced mathematical study. The syllabus that follows is detailed and ready for academic use.

Contents

1. Quick Info

- **Lectures:** 2 per week (90 min)
- **Problem Session:** 1 per week (120 min)
- **Assignments:** Weekly
- **Quizzes:** Monthly
- **Assessment:** Assignments and Quizzes
- **Recommended Level:** Undergraduate (1st–2nd year)
- **Prerequisites:** Basic calculus and comfort with proofs
- **Software:** Python (SymPy), R, LaTeX

2. Course Overview

This course covers:

- Syntax and semantics of propositional and predicate logic.
- Set theory basics: operations, relations, functions, cardinality.
- Core combinatorics: counting, bijection, inclusion–exclusion, and generating functions.
- Interplay between logic, sets, and counting.
- Practice in problem-solving and proof-writing.

3. Learning Outcomes

After completing this course, students will be able to:

1. Translate natural-language statements to formal logical expressions and vice versa.
2. Construct and critique direct, contrapositive, and contradiction proofs.
3. Manipulate set expressions and reason about functions, relations, and cardinality.
4. Solve combinatorial problems using bijective and analytic methods.
5. Apply logical and set-theoretic ideas to discrete structures and algorithms.

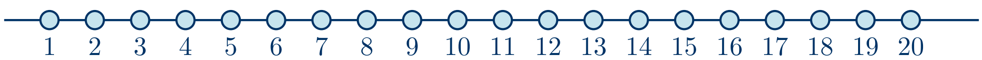
4. Pedagogy and Teaching Methods

- **Interactive Lectures:** argument building and conceptual examples.
- **Problem-Solving:** weekly structured problem solving with graded difficulty.
- **Proof Clinics:** focused sessions on writing and polishing proofs.
- **Assessment for Learning:** feedback-first grading on assignments.

5. Detailed Week-by-Week Plan (20 weeks)

Week	Topics and Activities	Focus / Deliverables
1	Introduction to Mathematical Reasoning; propositions, connectives; truth tables; basic proofs.	Logical Fundamentals, Proof Warm-up
2	Logical equivalences; normal forms; implications; proof techniques (direct).	Assignment 1: Truth Tables + Short Proofs
3	Quantifiers, predicate logic, negation of quantified statements, common pitfalls.	Assignment 2: Quantifier Exercises
4	Sets: notation, union/intersection, basic identities; Venn diagrams.	Problem Sheet 1: Set Identities
5	Functions, injective/surjective/bijective; composition; inverse functions.	Assignment 3: Bijection Constructions
6	Relations, equivalence relations, partitions; partial orders.	Workshop: Relations & Partitions
7	Cardinality: finite, countable, uncountable sets; Cantor diagonal argument.	Quiz 1 + Cantor Argument Practice
8	Logic meets sets: characteristic functions; Boolean algebra; De Morgan's laws.	Comparative Proof Assignment
9	Proof writing clinic: contrapositive, contradiction, induction refresher.	Proof Portfolio Update
10	Basic combinatorics: rule of sum/product; permutations, combinations.	Assignment 4: Counting Problems
11	Binomial coefficients, combinatorial identities, Pascal's triangle.	Practice: Combinatorial Identities
12	Inclusion–exclusion principle; Pigeonhole principle.	Assignment 5: PIE Applications

Week	Topics and Activities	Focus / Deliverables
13	Combinatorial proofs; bijective proofs; generating functions (intro).	Reflection: Proof Strategy Essay
14	Counting functions and relations; number of relations, number of functions.	Assignment 6: Counting Function Classes
15	Boolean functions, truth-table enumeration, and circuit perspective.	Coding Task: Logical Enumeration
16	Counting equivalence relations and partitions (Bell, Stirling numbers).	Quiz 2 + Project Discussion
17	Advanced interplay: quantifiers in combinatorial contexts; case analysis.	Reading: Logical Combinatorics
18	Applications: logic in CS, databases, complexity theory pointers.	Mini Presentation: Logic in CS
19	Final project presentations: Combinatorial Logic Integration.	Capstone Presentations
20	Revision, reflections, course survey, and feedback.	Final Reflection Submission



Timeline of Weeks (1–20)

Logic → Sets → Combinatorics → Integration → Assignments

6. Weekly Assignments and Sample Problems

Each weekly assignment will have a mix of:

- **Core problems** (proof-based, required)
- **Stretch problems** (challenging; optional for extra credit)
- **Short reflections:** “Explain in 150 words the reasoning strategy you used”

6.1. Sample Problems

1. Prove: If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.
2. Show that the set of rational numbers is countable.
3. Use inclusion–exclusion to count the number of onto functions from a 5-element set to a 3-element set.
4. Provide a combinatorial proof of $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.
5. Prove or disprove: The union of two countable sets is countable.

6.2. Final Assignments

An assignment integrating logic, sets, and combinatorics:

- **Counting Logical Outcomes:** enumerating distinct logical formulas under equivalence.
- **Set-Theoretic Models of Boolean Expressions:** mapping circuits to set operations.
- **Combinatorics of Equivalence Relations:** counting partitions with constraints.

Deliverables: a 6–8 page report (LaTeX), 12–15 minute presentation, and code appendix.

7. Software, Tools, and Reproducible Assignments

- Python (`itertools`, `sympy`) for enumerations.
- R for combinatorial simulations.
- GitHub Classroom for assignments.
- LaTeX templates for reports.

8. Primary Texts and References

- Daniel J. Velleman, *How to Prove It: A Structured Approach*
- Paul R. Halmos, *Naive Set Theory*
- Kenneth H. Rosen, *Discrete Mathematics and Its Applications*

8.1. Additional References

- Herbert B. Enderton, *Elements of Set Theory*
- Susanna S. Epp, *Discrete Mathematics with Applications*
- Peter J. Cameron, *Combinatorics: Topics, Techniques, Algorithms*
- Elliott Mendelson, *Introduction to Mathematical Logic*
- Richard Hammack, *Book of Proof (free online)*